PHYS 580 Homework 1 (Chap.1,2)

All questions are worth 10 points, irrespective of complexity or length. Please submit all your solutions on Brightspace in PDF format, together with the source code (if any). Make sure to

a) discuss the physics and your results, as appropriate and/or directed,
b) include graphical output to illustrate the results,
c) include the source codes of your programs (at least the critical parts thereof),
d) describe briefly what your program does, and how it does it (algorithm), and
e) state the nature of the numerical approximation you used and demonstrate how you know that the approximation (with the parameters you used) is adequate for the problem you used it for.

Note the errata pages for the textbook: http://www.physics.purdue.edu/~hisao/book/www/errata.pdf
which are explained further on http://www.physics.purdue.edu/~isao/book/www/errata.html

1. Problem 1.4 from the Giordano-Nakanishi book (p.16), with $\tau_A/\tau_B = 1/3$, 1, and 3. Make sure to include a discussion of any relationships among the time scales (τ_A, τ_B) and the time step you chose for the numerical work. Explore and interpret the results for various initial conditions such as $N_A(0)/\tau_A > N_B(0)/\tau_B$, or $N_A(0)/\tau_A < N_B(0)/\tau_B$.
Note, if your analytic result does not "like" $\tau_A = \tau_B$, then you have to set $\tau_B = \tau_A + \varepsilon$ in it and then take the $\varepsilon \rightarrow 0$ limit. (If the limit is giving you trouble, just shift τ_B by a very small amount).
Optional: How would the differential equations change if B becomes A when it decays?

2. Problem 2.2 (p.24).
Investigate the effect of varying both the rider's power and frontal area on the ultimate velocity. In particular, for a rider in the middle of a pack, the effective frontal area is about 30 percent less than for a rider at the front. How much less energy does a rider in the pack expand than does one at the front, assuming they both move at a velocity of 13 m/s?

3. Problem 2.7 (p.31).
Use the *adiabatic* model of the air density (2.24) to calculate the cannon shell trajectory, and compare with the results found using (2.23). Also, one can further incorporate the effects of the variation of the ground temperature (seasonal changes) by replacing B_2 by $B_2^{ref}(T_0/T_{ref})^{\alpha}$, where B_2^{ref} is the value of B_2 at a reference temperature T_{ref} and T_0 is the actual ground temperature. The value quoted in the text is appropriate for $T = 300$ K. In particular, how much effect will the adiabatic model have on the maximum range and the launch angle to achieve it? How much do they vary from a cold day in winter to a hot summer day?

4. In all calculations of cannon shots so far, we neglected the fact that the projectiles are launched from and measured in the rotating reference frame of Earth. Taking rotation into account would add a term $-2\vec{\omega} \times \vec{v}$ to the apparent acceleration in Earth's frame of reference (due to the Coriolis force), making even the spinless cannon problem 3-dimensional. Estimate the effect of the Coriolis force on the trajectory of a typical cannonball launched

toward southeast from Lafayette (latitude 40º25’ N) with $v_0 = 700$ m/s at $\theta = 45$ degrees with respect to the horizontal.

5. (Order of magnitude checks.) On p.28 of the Giordano-Nakanishi book, the air drag coefficient for a large cannon shell is said to be $B_2/m \approx 4 \times 10^{-5}\ \mathrm{m}^{-1}$. On p.38, the magnitude of the Magnus term in baseball is stated to be $S_0/m \approx 4.1 \times 10^{-4}$. Furthermore, p.45 gives an estimate $S_0\omega/m \approx 0.25\ \mathrm{s}^{-1}$ for the golf ball, and the next page (p.46, Problem 2.24) says that for a ping-pong ball $S_0/m \approx 0.040$. Argue about the orders of magnitude of these values and justify them if you can. If needed, refer to the official specifications for the various balls (see, e.g., the document HW1-BallSpecs.pdf posted in the Homework area on Brightspace). If you think that any of the above numbers in the text are unreasonable, then state why that is so.